

CHAPTER 1

A PLAN OF ATTACK

MAYHEM

At first glance racing looks like mayhem; your first ride in a car on a racetrack even more so. The straights go by in a blur as the car bounces around and the wind blasts past the open top. As you approach the end of the straight your instinct screams that there is *no possible way* the driver's going to make this turn unless he brakes *now*!



Fig. 1-1. A quick tour of the Sebring 12 Hour course with '96 Champion Jim Pace. "What just happened?!"

There are three basic problems to solve in racing: 1) driving on the best path, 2) carrying speed away from corners onto straights, and 3) efficiently slowing the car at the entry to corners.

But the driver continues on, foot to the floor on the throttle, unconcerned. You briefly wonder if he has had a coronary when *wham*, you get slammed forward on the

belts as the driver applies the brakes and a second later thrown sideways as the car is committed to the corner. *It's sliding, it's sliding!* you scream to yourself, convinced the driver has lost control. But no, a few twists of the steering wheel and some magical inputs on the pedals and you scream off onto the next straight. This happens again and again as you proceed around the course. When it's finally over, you climb out of the car slightly limp, exhilarated, and not quite sure what just happened.

It seems like chaos because the novice isn't clued in to the thinking and planning that is going on inside the helmet of the driver. To have any chance of doing well at the task of driving a racecar, you have to take it out of the realm of mystery and bring it down to earth. You need to develop a clear idea of what you need to do, and why, at every point around the racetrack. It is also important to avoid blind alleys, to find out which aspects of driving hold the biggest potential reward in terms of speed, and which skills lead to smaller gains.

LINE, CORNER EXIT SPEED, BRAKING

We have found that in the quest for fast laps there are three main areas that you need to focus on to reap the most rewards on the racetrack. The first is to find the most advantageous path around the racetrack. You'll find that where you drive the car has a profound impact on both cornering speed and straightaway speed.

The second priority is to learn the skills required to drive on this best path, getting closer and closer to the limits of the car's ability, and carrying optimum speed through the corner, onto the following straight.

The third area is to develop skill in slowing the car from straightaway speed to the appropriate speed for the upcoming corner.

In this first chapter you will learn why this three-tiered approach works, and you'll develop a clear understanding of how a talented race driver juggles a car's ability to accelerate, brake, and corner in order to minimize lap time.

DEFINING THE PROBLEM

To start this process, let's first define the fundamental goal of a racer: Confronted with a twisty piece of asphalt, you need to figure out how to drive around it in the shortest possible time. That's it. It's a problem of *minimizing time*.

A Basic Racetrack

Figure 1-2 depicts the simplest possible road circuit, made up of six corners, five rights and one left. To start to figure out how to minimize time, let's divide the track into sections: straightaways and corners.

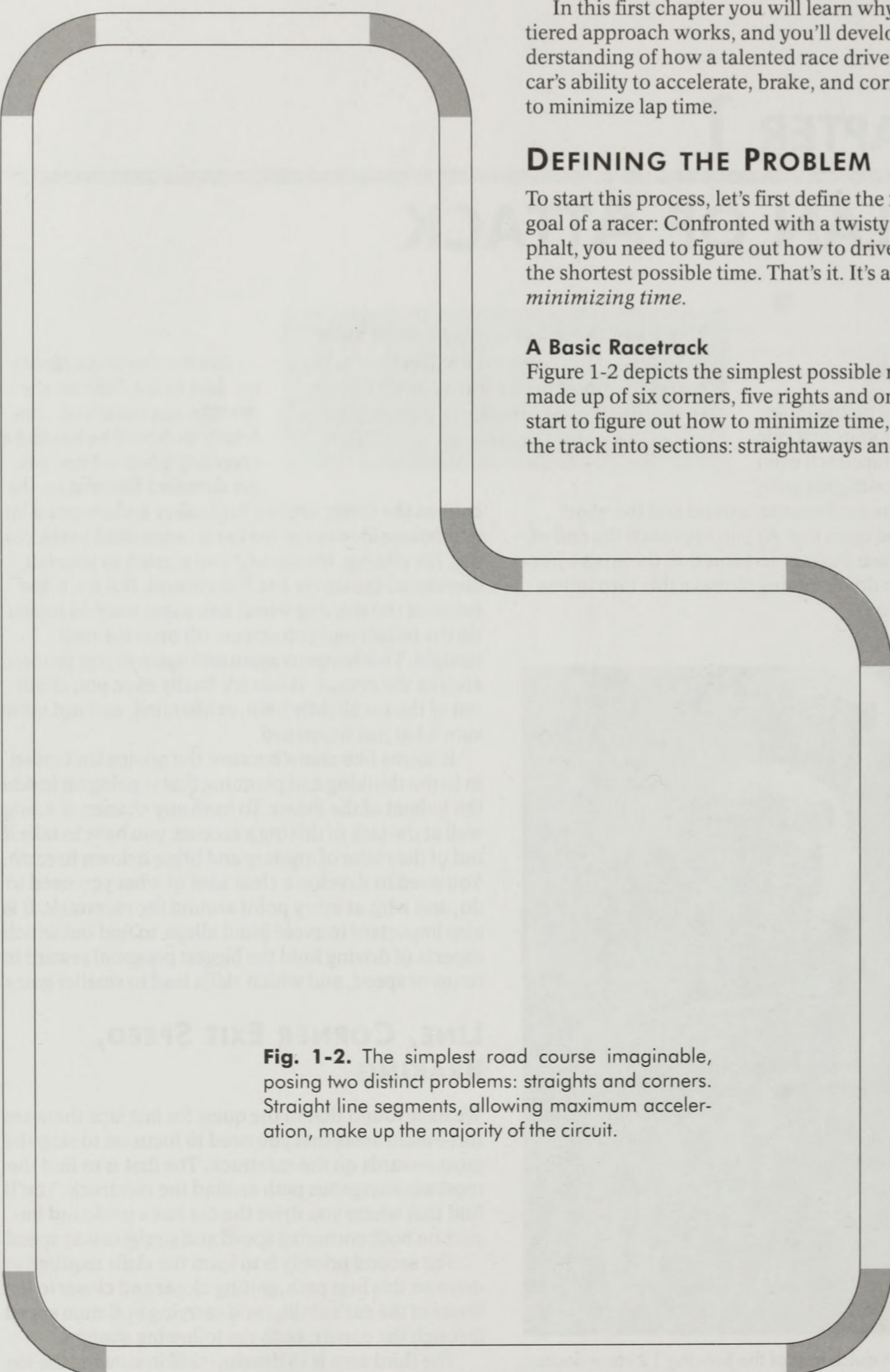


Fig. 1-2. The simplest road course imaginable, posing two distinct problems: straights and corners. Straight line segments, allowing maximum acceleration, make up the majority of the circuit.

Straightaways

How do you drive down the straight in the minimum amount of time? *As fast as possible.* (The questions *do* get harder.)

In all seriousness, you minimize time spent on the straight by accelerating as hard as possible, right foot to the floor on the accelerator, every moment the car is on the straight. You want to use all the horsepower the motor can develop to accelerate the car's mass.

In Fig. 1-2, the unshaded portions of our track map are the areas where you'll do this; and you'll discover that this area covers 70% to 80% of the racetrack, an important point to remember. A racecar typically spends much more time on straightaways at maximum acceleration than anywhere else.

Sadly, there are complications, and they crop up rather quickly (see Fig. 1-3).

Let's say that our racecar has sufficient horsepower to gain 100 m.p.h. between the beginning of the main straight and the corner at the end of the straight. This means that even if you drive onto the straight at 10 m.p.h., the car will end up at Turn 1 going 110 m.p.h. Now, there's no way you are going to make Turn 1 at 110 m.p.h.—Michael Schumacher himself couldn't drive faster than 55 through Turn 1 with our car.

So, somewhere toward the end of the straight you'll have to use the brakes to slow the car down to around 55 m.p.h. in order to turn right and continue your lap rather than go off into the weeds.

Now your straightaway looks like an acceleration zone for most of its length, with a deceleration zone, or braking zone, at the end, represented in Fig. 1-3 by Xs. For now, let's leave the discussion of the braking zone until later. First, you need to figure out how to spend as little time as possible on the straight getting to the braking zone.

If you devised some way to drive through the corner leading onto the straight faster than 10 m.p.h.—say, 20 m.p.h.—it's obvious that you would be faster through the corner. More importantly, however, you would end up at the end of the straight going 120, thus the elapsed time down the straight would be lower.

Corners Are Part of Straights

Now you can see that the average speed down the straight is directly affected by the speed through the corner leading *onto* the straight. So, to minimize the time spent on the straight, you need to work on increasing your speed through the corner.

It's easy to go faster at low speeds—you just use a little more throttle and the car drives easily through the corner 10 m.p.h. faster. Eventually, as you try to go faster and faster, the car will have to work harder and harder to change direction.

We've already touched upon using the car at its maximum, or its "limit," when we said you were going to accelerate down the straights using 100% of its

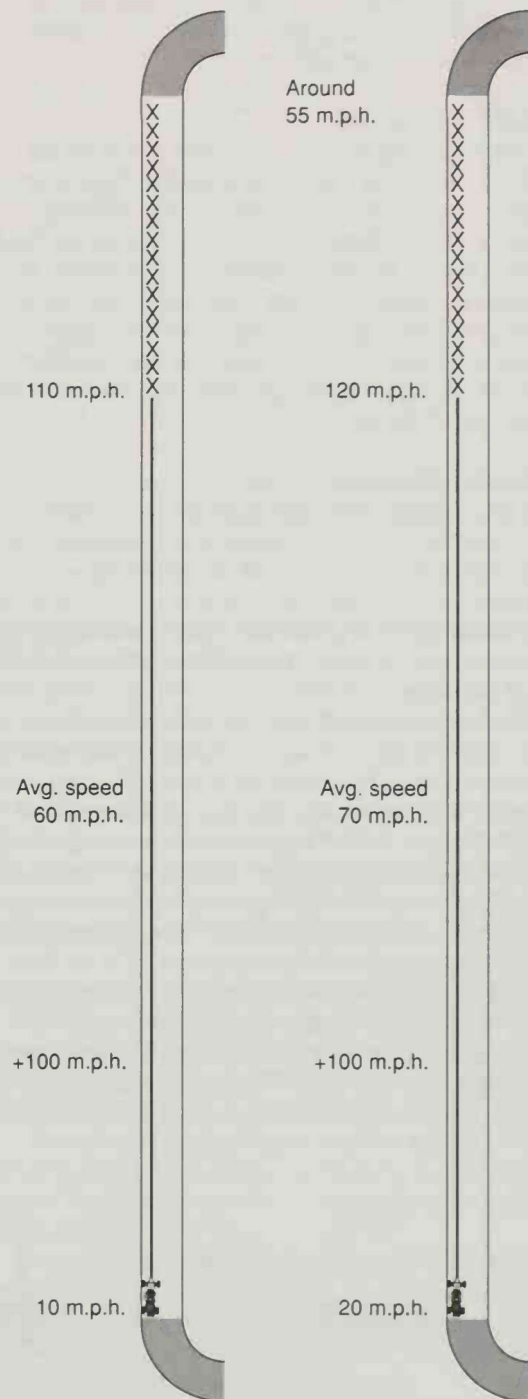


Fig. 1-3. Accelerating as hard as you could, using all the horsepower available, this car would gain 100 m.p.h. from the exit of the turn to the point at which braking would have to begin. Changing only the speed at which you exit the previous corner, the average straightaway speed increases 10 m.p.h.

ability to accelerate. To minimize the time it takes to drive through a corner, you would like to be able to use 100% of the car's ability to turn.

"Limit" Cornering

If a car is moving and you take your hands off the steering wheel, the car will go straight. If you want to change the car's direction, you turn the steering wheel. This uses the grip of the tires against the road to force the car to change direction. Remember, it wants to go straight, and the tires are *forcing* it to turn. When going slowly, or turning the wheel slightly, it doesn't take much force to make the car turn. But raise the speed, or tighten the turn, and the tires will need to work harder.

Using the Skidpad

In your everyday driving you seldom use more than 40% of the tires' ability to work. If you want to experiment with how tires behave at their cornering limits, you need to be in a safer environment where there are no oncoming cars or giant oaks nearby, waiting to turn a mistake into a disaster. The public roads are definitely *not* the place to do this.

We use a skidpad, which is a fancy name for a wide-open flat piece of asphalt devoid of any solid objects to run into. You could use a specially constructed pad at a racetrack or a big empty parking lot.

Take the car to the "pad," turn the steering wheel 45 degrees (right or left), and drive away—slowly at first. If you keep the steering wheel in that one position the car will travel in a circle. Now gradually increase your speed.

You can't go faster indefinitely. At some point, you'll be using *all* the grip the tires have—100% of their cornering traction. At this top speed, the tires are working terrifically hard. They're screaming, literally sliding over the surface of the road. If you're driving a

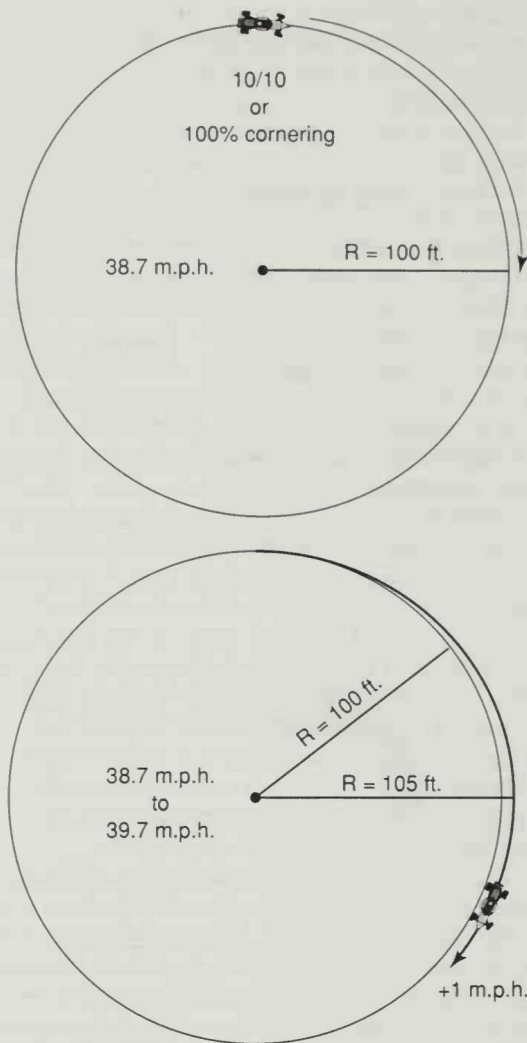


Fig. 1-5. Once the tires are at their cornering limits an increase in speed results in an increase in radius.

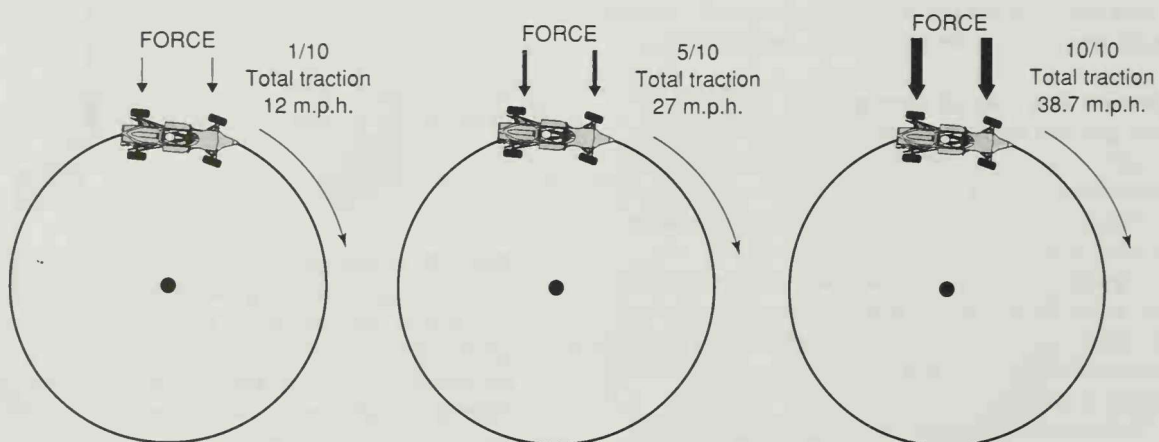


Fig. 1-4. As a car travels faster and faster around a circular path, the tires use increasing amounts of their potential traction to force the car to continue to change direction.

street car, it will have rolled substantially toward the outside of the turn and your body will be pushed to the outside of the circle with a force equal to at least 75% of your body weight. The car is at its cornering "limit" using 100% of its cornering ability.

If, at this point, you add more speed, surprisingly enough nothing terrible happens. The car continues to turn, but it turns on a bigger arc (see Fig. 1-5). At the limit of a car's cornering ability, the speed it can attain is directly related to the radius of the arc it's on. So, in cornering, if you want more speed, drive on an arc with the biggest possible radius. Sounds simple, doesn't it? Yet, it's amazing how many active racers fail to make use of this basic law of physics as it applies to racecars.

All of this information applies to our quest to minimize time around the racetrack. So far we've concluded that:

- To minimize time on the straight, use 100% acceleration on the straight—wheel straight, flat out;
- You need to exit the corner leading onto the straight as fast as possible;
- In cornering, the biggest radius yields the highest speed.

FINDING THE LINE

If you want to drive through a corner at the highest possible speed, then you need to find a path through the corner on an arc with the biggest radius. You need to find *the line*.

Driving up to the corner, you have a lot of choices about how to drive through it (see Fig. 1-6). You can drive around the inside of the corner, but if you do you're restricted to the smallest radius, R1. You can make the radius through the corner bigger by driving around the outside of the corner—the radius R2 is bigger than R1, so the maximum speed *will* be higher.

The best bet, however, is to start the arc before the corner, touch the inside edge of the road halfway through, and touch the outside edge of the road again at the exit. If you do this, you can use radius R3, clearly the biggest and best choice.

Why "The Line" is Step 1

When you drive a car at its cornering limits, it's the line you choose (the radius of the arc the car is on) that determines maximum speed through a corner. And the speed you carry out of the corner determines maximum speed on the following straightaway. Since the line ultimately determines both cornering *and* straightaway speed, it's obviously the most important place to start.

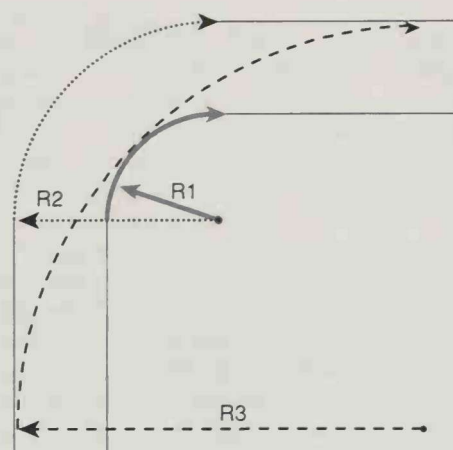


Fig. 1-6. You can choose various paths through a corner, but only one yields the biggest radius: "the line."

Remember, the process of finding the optimum line—the best combination of arcs through corners and paths down the straights—is a tool to use to put yourself in a position to post the fastest possible time around the racetrack.

Charting a Lap

Let's take another look at our typical track map to see what you're going to do with the controls in order to get around the place (see Fig. 1-7). You might not get all this exactly right on the first try so, at the start, you won't try to go as fast as you know you ultimately will when, through practice, you earn the right to go faster.

Coming out of Turn 6 onto the main straight (the longest), you'll accelerate in a straight line toward Turn 1. Again, the acceleration zone is designated by a solid line.

Somewhere toward the end of the straight you'll use the brakes to slow the car down to an appropriate speed for the next corner. The braking zone is designated by a series of Xs.

At a point on the left-hand side of the straight you'll turn the steering wheel to the right, initiating an arc that touches the inside edge of Turn 1 and finishes at the outside edge of the road at the exit. The cornering zone is designated by double lines.

Exiting Turn 1 and heading toward Turn 2, you'll repeat the same routine. You can continue to follow the car's progress around the course, referring to the coding to see which phase of the car's ability is being used at any particular point on the track.

As you can see from this simplification of your lap, a car has only three fundamental abilities: 1) Acceleration, 2) Braking, 3) Cornering.

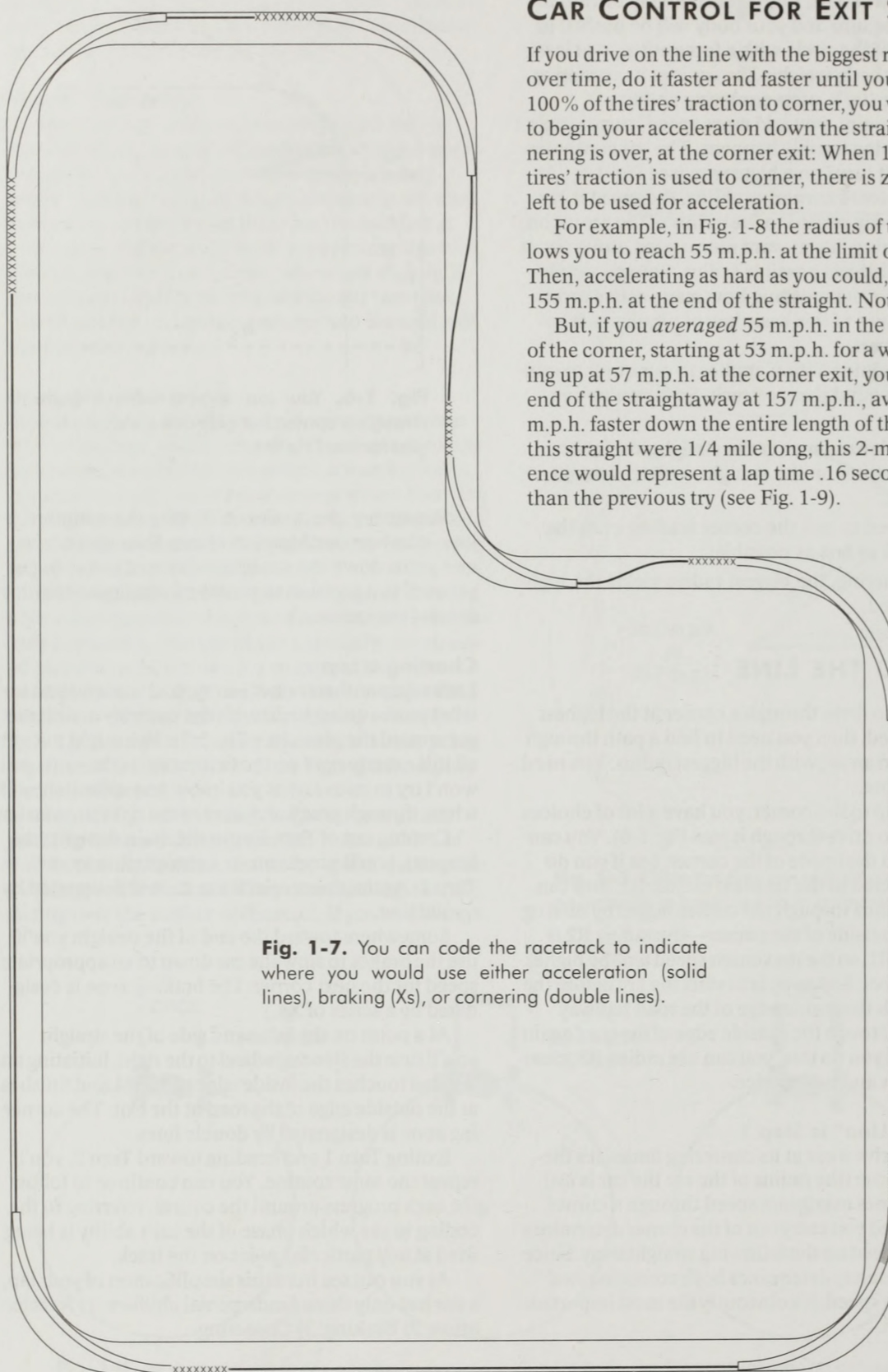
CAR CONTROL FOR EXIT SPEED

If you drive on the line with the biggest radius and, over time, do it faster and faster until you're using 100% of the tires' traction to corner, you won't be able to begin your acceleration down the straight until cornering is over, at the corner exit: When 100% of the tires' traction is used to corner, there is zero traction left to be used for acceleration.

For example, in Fig. 1-8 the radius of the corner allows you to reach 55 m.p.h. at the limit of cornering. Then, accelerating as hard as you could, you'd reach 155 m.p.h. at the end of the straight. Not bad.

But, if you *averaged* 55 m.p.h. in the second part of the corner, starting at 53 m.p.h. for a while but ending up at 57 m.p.h. at the corner exit, you'd reach the end of the straightaway at 157 m.p.h., averaging 2 m.p.h. faster down the entire length of the straight. If this straight were 1/4 mile long, this 2-m.p.h. difference would represent a lap time .16 seconds lower than the previous try (see Fig. 1-9).

Fig. 1-7. You can code the racetrack to indicate where you would use either acceleration (solid lines), braking (Xs), or cornering (double lines).



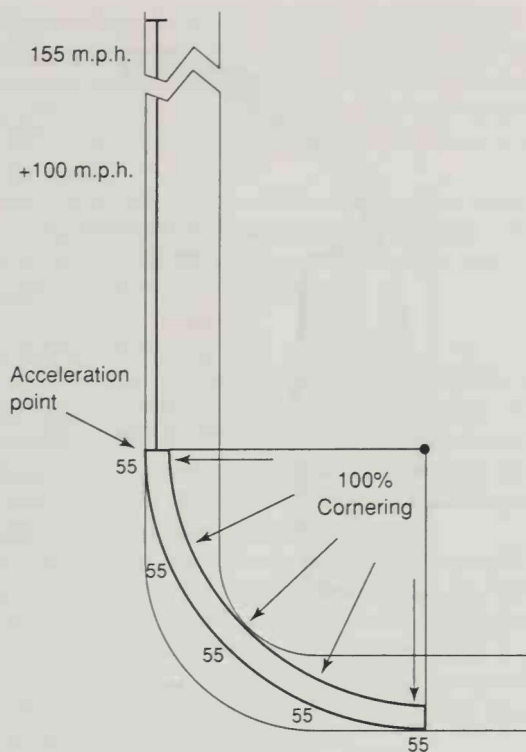


Fig. 1-8. If you corner at 100% from the beginning to the end of the turn, you can't begin accelerating until you complete cornering.

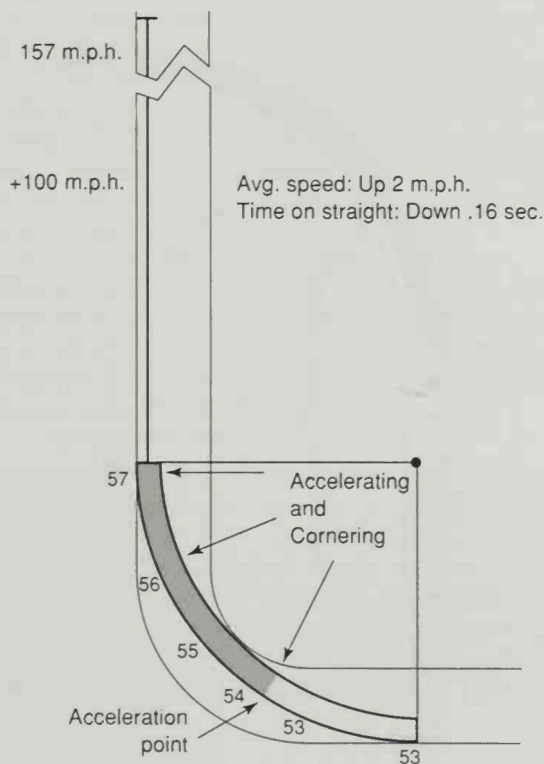


Fig. 1-9. If you combine acceleration and cornering, the higher speed at the exit of the corner reduces the time it takes to drive down the straight.

Mixing Abilities

We've so far described "the limit" as using the tires' grip in *either* acceleration, cornering or braking. In reality, you can combine these abilities. If you decide to use some acceleration while cornering, the car will do it, with a compromise. If you want to use 20% of the tires' traction for acceleration, you only get 80% of the traction for cornering.

In coding the racetrack, you now need another icon to represent a mix of acceleration and cornering. Starting with Fig. 1-9 we're using a shaded area.

This loss in cornering traction affects your path through the corner: the car will still change direction but, because of the reduction in turning force, it will change direction on a bigger radius arc than before. Additionally, as you accelerate away from a turn, the speed is getting higher and, as you know from your skid pad experience, the higher the speed, the bigger the radius required (see Fig. 1-10).

So, as you're accelerating and cornering you need to increase the radius of the arc you're on. You do this by gradually unwinding the steering wheel as you exit the turn.

Since the greatest part of a lap is spent on corner exits and straights, any speed improvements on this portion of the course, which typically represents 70% to 80% of a lap, would have the greatest effect on decreasing lap time. This strategy takes the focus off

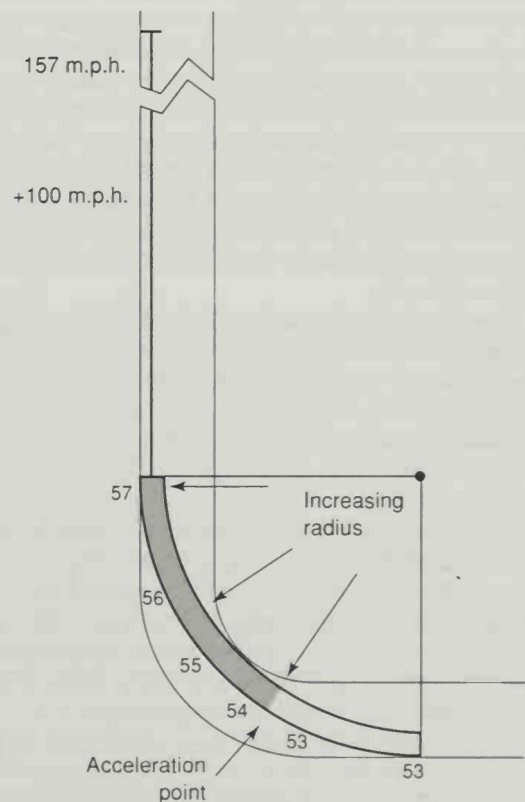


Fig. 1-10. As the cornering speed increases, the arc's radius at the corner exit must also increase.

braking: don't be tempted, as many drivers are, to go immediately to late-braking when you decide that it's time to go faster. It's much more important to get to your throttle application point at the appropriate speed and experiment with driving the car faster and faster through the last half of the corner, under power.

The Order of Effort

Both the line and corner exit speed are extremely important, and just as important is their order of priority. You need to first determine the correct path on which to drive and *then* work on increasing your speed on that path.

Learning the line is easier than many of the other things you'll have to learn to do in a racecar, so indulge yourself at the beginning and concentrate on getting the car spot-on the best path. It's exhilarating to learn the car control skills required to maximize your exit speed, but you have to be careful to not let the kick of sliding the car around get in the way of your ultimate goal: lower lap times.

SLOWING FOR CORNERS

The third basic skill is learning how to slow the car for a corner and how to drive it from where the turn begins to the throttle application point. We call this phase of driving racecars "braking and entering."

Let's go back and consider Turn 1 on our hypothetical racetrack. You exited the corner onto the straight accelerating through 57 m.p.h.; down the following straightaway, the car gained 100 m.p.h. so that you were approaching Turn 1 at 157 m.p.h. At some point on the approach to Turn 1, you'll feel the need to slow down for the rapidly approaching corner. You're going straight here, so you can use 100% of the car's potential braking ability. We call it "threshold braking": threshold, because to get to the 100% level of stopping power, you're at the threshold of wheel lockup, right on the edge.

It seems obvious that slowing the car in a straight line using 100% of its braking ability will reduce the speed of the car to the level you want it in the shortest possible time and distance. No argument. However, let's take a closer look at this situation.

Brake and Turn

Let's say that Turn 1 is just like the corner that leads onto the straight: that is, its cornering limit speeds are in the neighborhood of 55 m.p.h. You could use the straight to slow the car from 157 to 53 m.p.h. by the time you turn the wheel into the corner. In fact, for decades the prevailing thinking was that any time the car was braking, it *had* to be going straight.

The truth, however, is that a car will *decelerate* and turn just like it's able to accelerate and turn. But, just like mixing acceleration and turning, braking and turning requires that you make some compromises.

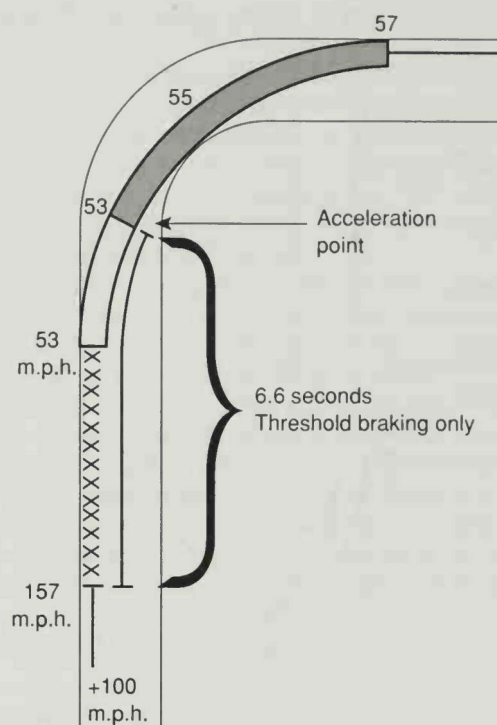


Fig. 1-11. Slowing the car only in a straight line requires that you drive at a constant speed from the point where you begin to turn up to the point where you begin to mix acceleration and cornering.

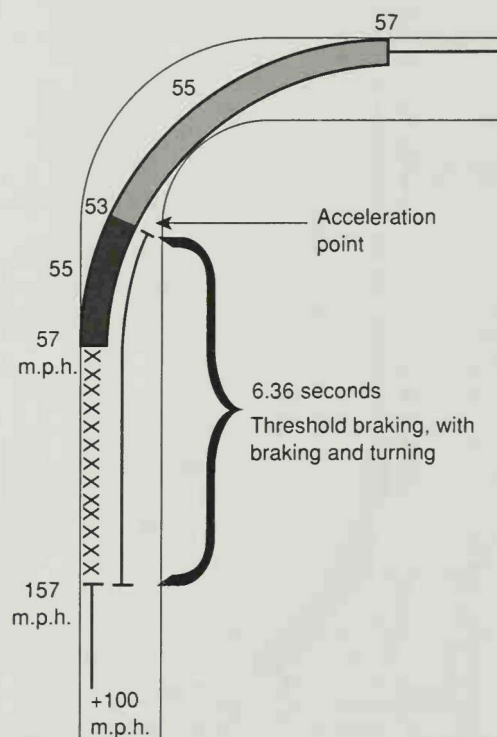


Fig. 1-12. By continuing to decelerate as you turn into the corner, your speed at corner entry can be higher.

Under threshold braking you're using 100% of the traction in the tires to brake. If, while threshold braking, you turn the steering wheel and ask for cornering force, the front wheels will point toward the corner, but the car won't change direction because there is no traction left for cornering.

But, if you reduce some of the braking effort, say, to 80% braking, you get 20% of the tires' traction available for cornering. With the steering wheel turned, that 20% bends the car into the turn.

Gains from Braking and Turning

We have already concluded that you're going to mix acceleration and cornering at the exit of the turn, by starting at 53 at your throttle application point and ending up at 57 at the exit—averaging 55 m.p.h. So the basic aim is to slow the car from 157 to 53 m.p.h. by the throttle application point.

If you restrict yourself to braking on the straight, you'd have to slow the car to 53 by the time you turn the steering wheel into the corner. You'd then drive from the turning point to the acceleration point at a constant 53 m.p.h. In a car with the braking capabilities of our school Formula cars, that would take about 6.6 seconds (see Fig. 1-11).

By continuing to brake and turn into the corner, starting at 57 at the turn-in and continuing to decelerate to 53 at the acceleration point (represented by the darker area in Fig. 1-12), you can average 55 across the same stretch of track, rather than 53 when you were only braking on a straight line. The time it would take you is about 6.36 seconds—.24 seconds less.

The lap time, by the way, is not insignificant. While a difference of .24 seconds may not seem like a lot on first glance, if you gain this advantage in four corners you're talking about .92 seconds per lap. In a 20 lap race this gain represents 18.4 seconds—and 20 laps is a short race.

Are these tenths of a second worth the effort? Maybe not at the beginning of a career when you are ten seconds off the pace. Using the braking and turning ability of the car won't get you the ten seconds, but, as you'll see in later chapters, lap time is just one of many reasons to develop this skill.

"I think that experimenting with your braking is very important, but after you get your line and after you get comfortable with the car coming off the corner. The last area is braking where you're going for that last little bit."

—Danny Sullivan

Charting a Lap, Take Two

Let's take an updated look at how to proceed around the entire racetrack using our latest code scheme (see Fig. 1-13).

On the straight, you'll use flat out 100% acceleration, designated by the solid line. When you reach the braking zone, designated by Xs, you'll use the maximum, 100% threshold braking. You'll decelerate into the corner, mixing the car's braking and cornering force, designated by the darkly-shaded area on our map. When you accelerate out, you'll mix the car's ability to accelerate with its cornering abilities, shown on our map as a lightly-shaded area. When you reach the next straight, you'll start the process over again, using 100% of the car's acceleration power.

Our coding of this hypothetical racetrack is, admittedly, very general at this stage, leaving out considerable detail. It's not meant to be a 100% accurate representation of exactly what you do on every inch of the course—rather, it's meant to get you thinking about what controls you are likely to be using as you lap the circuit.

OUT ON THE RACETRACK

If your only point of reference is driving on public roads, your first sense of excitement about driving on the track is that you can drive as fast as you like. You can roar down the straight at 90 m.p.h. and nobody complains. At first it's exciting to keep your foot to the floor as your car goes 30 to 40 to 50 m.p.h. faster than the legal speed limit. In fact the most common question we get about our school racecars is, "How fast will it go?" The surprise is that straight line speed is not all it's cracked up to be. It doesn't take much talent to hold the steering wheel straight and press the gas pedal to the floor, so there's no challenge in it. The speed is invigorating at first, but the thrill wears off quickly as you get accustomed to the scenery whizzing by.

The real excitement comes at the end of the straightaway where you need to slow the car down to negotiate a corner. This is where the newcomer to racing, all flushed with excitement from the straightaway speed, starts to fall apart. There are a lot of questions to be answered. Where do I put on the brakes? How long? How hard? What gear should I use for the corner? How and where do I downshift? Where should I steer into the turn? How fast can I go through it? Will the car spin? Will I crash?

Just *driving* around the racetrack, as opposed to a *lap record pace* around the racetrack is an entirely different way of covering ground in an automobile. You can start driving around a racetrack in your street car without much training or knowledge of racing technique as long as you use the same pace that you use in your street driving. You should have no trouble judging when to brake or when to shift.

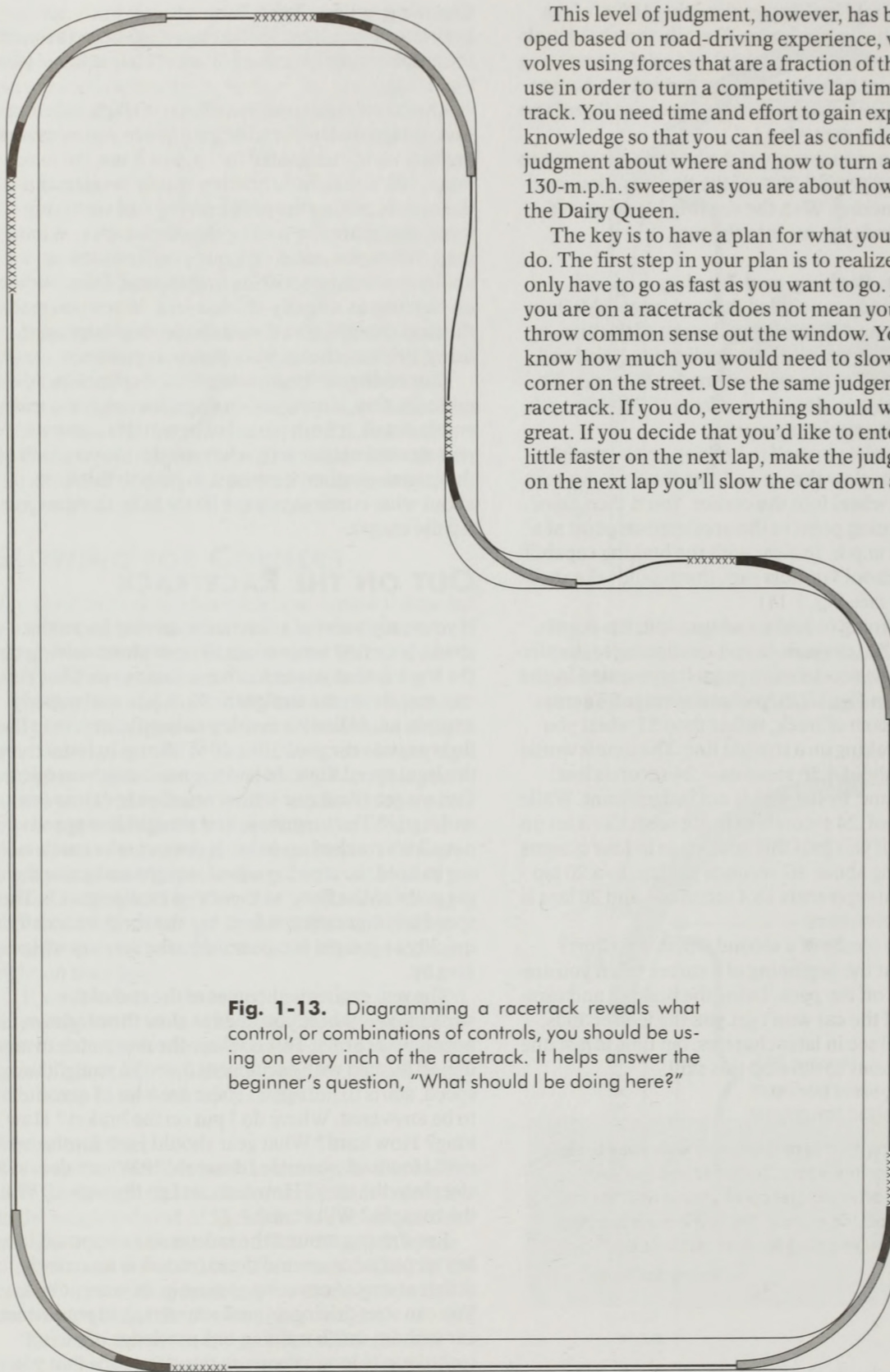


Fig. 1-13. Diagramming a racetrack reveals what control, or combinations of controls, you should be using on every inch of the racetrack. It helps answer the beginner's question, "What should I be doing here?"

This level of judgment, however, has been developed based on road-driving experience, which involves using forces that are a fraction of the forces you use in order to turn a competitive lap time on a racetrack. You need time and effort to gain experience and knowledge so that you can feel as confident making a judgment about where and how to turn a car into a 130-m.p.h. sweeper as you are about how to turn into the Dairy Queen.

The key is to have a plan for what you are going to do. The first step in your plan is to realize that you only have to go as fast as you want to go. Just because you are on a racetrack does not mean you should throw common sense out the window. You already know how much you would need to slow down for a corner on the street. Use the same judgement on the racetrack. If you do, everything should work out great. If you decide that you'd like to enter the turn a little faster on the next lap, make the judgement that on the next lap you'll slow the car down a little less.



Fig. 1-14. Spinning is never good, but once you know the common causes you can avoid them, and help take the anxiety out of approaching the car's limits.

"There's reaction time involved in everything you do in sports, but I think one of the reasons race drivers can continue to be good at older ages than other athletes is that experience has so much to do with it. You don't need 'young legs,' you just have to anticipate what's going on and think about it, looking that one step ahead.

"In racing, like many other sports, you have to get to the point where everything comes to you in slow motion. If you tried to return a serve from Agassi or Sampras, you wouldn't even be able to get the racket into position before the ball would be by you. If everything is coming at you in what seems like a normal fashion, a slower motion, you can deal with it. Experience plays its role in getting to that point. That's why stepping stones are so critical—that you go through the process of getting the experience."

—Danny Sullivan

The key word is "little." You'll find that the best bet is to increase your speed a little at a time, not to make a giant leap on faith alone.

Frequently Asked Questions

Now, let's answer some of the questions you might have about driving a car around a racetrack.

Where do I drive the car? The line. Regardless of whether it's Midget racers in Indianapolis or Formula 1 racers at Monaco, drivers go from the outside of the road at the beginning of the corner to the inside of the road in the middle, then back to the outside again at the exit. When you're on the track you should try to

do the same. Turn the steering wheel smoothly; no jerking the wheel to-and-fro like you see in the movies. Expect that you won't do it perfectly on the first try, but keep at it.

What gear should I use? Try a gear and see how it works. You want to be driving through the corner with the car in a gear and your right foot on the throttle—how heavily on the throttle is up to your judgment. If the engine is at low revs and the car feels like it's bogging, try a lower gear next time. If you try the corner in a gear where it reaches the engine's redline RPM before the end of the corner, you'll have to try a higher gear next time to avoid hurting the engine.

How and where do I downshift? If you're driving a standard shift car, which you are likely to be doing on a racetrack, you're stuck with downshifting, like it or not. You'll likely be in a higher gear on the straightaway than the gear which will work best, at lower speed, in the corner. The aim is to downshift without upsetting the car by chirping the rear tires when the clutch comes out in the lower gear. This is as true in your street car as it is in an Indy Car, and it requires you to learn how to tap or "blip" the throttle while still having part of the right foot on the brake pedal. This "blip" just before the clutch comes out in the lower gear is something you'll have to learn how to do for the racetrack. It's a skill that will also make you a smoother driver on the street. Besides, it sounds zoomy.

Where to downshift is a much easier problem. If you put it in a lower gear too soon you might over-rev and damage the motor, so you'll have to wait until the car is going slow enough. On the other hand, you can't downshift too late. You need the car to be in the right gear by the time you decide to go back to the throttle to drive through the corner. Downshifting anywhere between these two points is okay.



Fig. 1-15. Reference points, like these braking markers at Sebring, replace guesswork as a means for going faster.

Will my reaction times be fast enough? While it's often impossible to improve reaction time, the good news is that racing is much more about *anticipation*. Sure, there are times when something unexpected happens and you have to react. But far more often you need to use your ability to consistently plan ahead and take some action at a specific place and time. Believing that reaction time is all-important tends to make you move the controls violently and sharply. Violent maneuvers on the brakes, steering wheel, gear lever, clutch or throttle are a hindrance to fast laps, not a help.

"I think having good reaction times can get you out of trouble in certain situations, but thinking ahead, having enough foresight to know what you and the car are going to do, allows you to avoid getting yourself in situations where you need quick reactions. I find in a good normal lap in an Indy Car that I'm never reacting. You're making adjustments and corrections but things aren't surprising you. You're not thinking, 'Whoa, I didn't think the car was going to do that.'"

—Bryan Herta

Will I spin or crash? You don't have to. One of the biggest fables in the sport is that you have to spin to know where the limit is. It's not true. By approaching

the limits of the car's capabilities cautiously, you will get plenty of warning that it is nearing a spin. Still, racing is different from street driving in that you're likely to operate a car closer to its limits on the racetrack. When you're close to the limit, you have to pay attention to the details. Actions which, in your everyday driving, may be just sloppy or choppy can mean big trouble on the track. Fortunately, the most common mistakes which cause race drivers to lose control fall into just a few major categories.

The Four Most Common Mistakes

1) Running Off the Road Exiting Corners. As you drive more, you'll find that once you turn into the corner, you hold the steering wheel at pretty much the same angle all the way through the turn. Instead of jerking the wheel to and fro, you should keep constant steering pressure on the wheel, slowly releasing this pressure as the car drives onto the straight.

At the approach to the corner, if you turn the steering wheel toward the inside of the road too early, it may force you to turn the steering wheel even tighter as you reach the end of the corner. If you're going slowly, the car will respond to this second turning motion; but if you're going faster, you could get to the point where the car won't respond, and it will run wide coming out of the corner regardless of what you do to persuade it not to.

You need to control your desire to go fast until you know exactly where you want the car to be on every inch of the racetrack—you have to know "the line"—before you can safely go fast.

2) *Going Hopelessly Late Before Braking For A Corner.* Most professional racers use an easily recognizable marker at the approach to a corner as a cue to start braking. Novices need "brake points" too. You can easily avoid going too late by starting with a conservative brake point and, if you like, move it in toward the corner in small increments.

"Let's say you're braking at the 300 mark with no problem. Do you move the next spot to the 200? If you make that, do you jump to the 100? No way. You've got to take small steps to find out where that limit is. My way of doing it is to go just past the 300, and the later I go, I'm going to start taking it in much smaller steps, because if I know I'm fairly close, I don't want to screw everything up by going impossibly late."

—Danny Sullivan

3) *Bad Downshifts.* Chirping the rear tires on a downshift as you turn into a corner can spin you around very quickly. Practice smooth downshifts in your everyday driving and make sure to downshift before entering the corner.

4) *Carelessly Lifting Off the Throttle.* When a car is cornering fast, its behavior depends on keeping a nice balance between the grip of the front and rear tires.

An abrupt snap off the gas upsets this balance and can steal traction from the rear tires, causing a spin. Small, smooth, and subtle changes of the controls are the best bet on the racetrack.

A SAMPLE RUN

Let's go wring out a quick piece of machinery to get an idea of what a good driver does with the controls in a typical run around a track. Our winter operations are run out of our facility in Sebring, Florida—100 yards from the famous 12 Hour course. For starters, let's run a Viper around part of the test circuit, the part where you'll experience the highest and the lowest speeds the car will attain on a lap (see Fig. 1-16).

As you exit Turn 8 and approach Turn 9 (Fig. 1-17), you're getting the car positioned on the extreme right side of the road so that you can form the gentlest arc possible through this long left-hander. On the approach you have slowed the car to a speed that you're comfortable with, about 65 m.p.h., and you've left the car in second gear.

At a precise point near the edge of the road you turn the steering wheel to the left, aiming for a point on the inside of the turn that you feel will put you on an arc that will end on the outside edge of the road at the exit off the corner (see Fig. 1-18).

After you turn the wheel, you gently relax the pressure you have been exerting on the brake pedal. As your right foot comes off the brake, it immediately moves over to the throttle and begins squeezing on the gas very lightly (see Fig. 1-19).

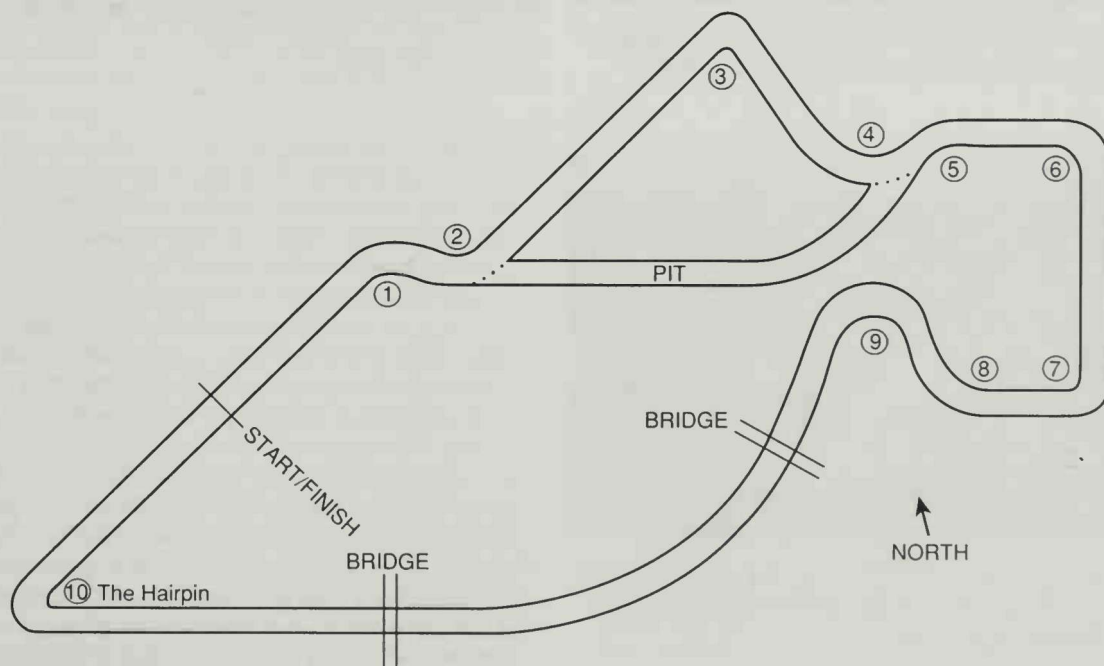


Fig. 1-16. Map of the Sebring Test Circuit, Sebring, Florida.



Fig. 1-17. Coming up to Turn 9, the "Carousel," you keep the Viper right, looking ahead for your turning point past the second curb.



Fig. 1-18. Relaxing the brake pedal, you transition over to the throttle and begin squeezing it on.



Fig. 1-19. The initial turn of the steering wheel hopefully puts the car on a path that touches the inside edge of the road.

Looking ahead, past the inside edge of the road, you're trying to judge whether the path to which you have committed is going to end up where you want, on the asphalt, but right on the outside edge of the road at the exit. You're trying to keep the car on a smooth arc so you're trying to avoid turning the steering wheel any more or any less than you did when you first turned into the corner.

Your eyes tell you that you're doing fine so you continue squeezing the throttle down past 30%, 40%, 50% of full throttle, concentrating on being smooth on the application and not being so greedy about it that you'll have to change your mind and reduce the throttle while still in the turn. The line for this lap looks good, so the throttle continues toward the floor, 60%, 70%, 80% (see Fig. 1-20).



Fig. 1-20. Exiting the corner, you continue to squeeze the throttle toward the floor as you relax the steering wheel.

Leaving the corner, you start to relax the amount that you have the steering wheel turned in order to smoothly exit the turn. You finish the corner on the right-hand edge of the road with the engine revs moving through 5300 RPM, around 65 m.p.h.

You have to shift at 5800 RPM and the V10 engine sounds great screaming up toward the redline. You have a half-mile of full throttle acceleration ahead of you and you're through 80 m.p.h. (see Fig. 1-21) as you shift to third gear and mat the accelerator once again.

The Sebring straight bends to the right for the first quarter-mile of its length and the Viper is gaining speed rapidly at full throttle. In many cars, this portion of the course is easy, but the Viper holds your attention. At 105 m.p.h. (see Fig. 1-22), you're swinging out to the left hand edge of the straight a few hundred feet before the pedestrian bridge, with its protective concrete wall six feet or so from the edge of the racetrack.



Fig. 1-21. The second to third shift at the bridge comes at 80 m.p.h.



Fig. 1-23. Your eyes are focused on the brake point, not the speedometer.



Fig. 1-22. Foot on the floor, the Viper sweeps to the left side of the road and the shift to fourth comes at 105 m.p.h.



Fig. 1-24. With the brakes on hard, the process of selecting the proper gear for the upcoming corner begins.

You're straight for an instant before shifting up to fourth here, so you peek at the speedometer one last time to see that you're at 110 m.p.h. with the car accelerating as strong as before in the new gear. You think for a second that there are two more gears to go—if only the straight were long enough.

The hairpin is coming up and you start looking for your brake point, foot still on the accelerator, squeezing it into the carpet. You're going faster than 125 m.p.h. but your eyes are outside, focused on the brake point, marked by the first of four pylons (see Fig. 1-23).

As you approach the first cone you quickly lift off the throttle and go immediately to the brake pedal, which you squeeze on hard (see Fig. 1-24). As the car slows in a straight line, you go down through every gear, blipping the throttle with the right half of your right foot just before the clutch is released in each downshift. Fourth to third, blip, clutch out. Third to second, blip, clutch out. You judge how much to blip

by the smoothness of the engagement of the next lowest gear. If the car leaps forward against the brakes, you blipped it too much. If the nose of the car dives lower as the rear tires try to speed the engine up, you blipped too little.

Next up, it's time to turn into the corner. The curb on the left-hand verge of the track has reflectors glued to the edge of the road for use at night in the 12 Hours; you know from experience that turning at the second pair of reflectors will put you on a path through the corner that touches the inside and outside edges—the gentlest arc through The Hairpin (see Fig. 1-25).

You turn at your reference point, the car already in second gear at about 35 m.p.h., ready to begin accelerating away from the corner. An instant after you turn the wheel right, you begin to relax the braking pressure, just as you did in Turn 9. From here on out of the corner it's the same routine—relax the last bit of brake pedal pressure, squeeze on the throttle, and relax the steering pressure as the car's path looks good.



Fig. 1-25. The focus now becomes the turn-in point, the second pair of reflectors glued to the track surface.

What is "Fast"?

It's remarkable how something so repetitive can be so much fun. As fast as it seems to the uninitiated observer, this run down to the Sebring hairpin was not fast in terms of the way a racer defines fast: time.

Through Turn 8 to the exit of the hairpin was done in 38 seconds. The total lap time was just over 1 minute, 30 seconds. The same car, driven closer to its limits, would be some four seconds per lap faster. It would be very hard to convince your first-time passenger that you could possibly go faster at all, let alone four seconds a lap, but the time is there to be gained.

Some drivers will choose to make the effort to get there. Some will be perfectly content to drive around the course at a slower pace, getting their enjoyment and satisfaction by going much faster than they would ever have a chance to in the highway environment. The techniques they would use to do this well and safely should be the same techniques that the driver on the lap-record pace uses. The primary difference between the two would be how closely each driver is willing to approach the limit of the car's ability, knowing that the closer one gets to the limit, the smaller the margin for error.

Bravery

We urge you all to not confuse skill with daring. Unfortunately, some auto enthusiasts are more willing to take risks with themselves and their cars than they are willing to exert the effort it takes to learn how to drive well. All drivers, especially drivers who make claims that they are better than the norm, have a responsibility to not endanger others in the pursuit of what is, after all, a sport.

We feel this responsibility keenly. Every week of the year we expose new enthusiasts to an area of speed and control they have never before explored.

"I think that the idea that you have to be fearless to be a good racer is the biggest misconception that anybody has about racing. I'm the world's biggest chicken. I am not brave at all. I've won a lot of championships and a lot of races over 25 years and I consider myself the biggest chicken out there. Sure, you can win some races on bravado but what looks to be bravery, later on, is knowing what you can and can't get away with. I would take one ounce of brains over two pounds of bravado."

—David Loring

As each school draws to a close, we hope that we have given the students skills that will have a positive effect. But beyond specific skills, the most important lesson is that in order to drive extraordinarily well, you have to use your head much more than your guts.

THE FRAMEWORK OF FAST LAPS

In this chapter, your goal has been to understand the framework in which you'll operate. You need to get a sense of which subjects related to driving racecars should be covered first so as to build a thorough understanding of what you'll try to accomplish in each of the areas that will help you turn fast, safe laps. The time-tested framework we use in our schools is to first concentrate on learning what it takes to turn a fast lap alone.

On the track as well as in the classroom, the first subject a racer needs to learn about is "the line," since it has such a great effect on both cornering speed and straightaway speed.

Big chunks of lap time will come off by learning to maximize the exit speeds coming off the corners, by mixing acceleration and cornering, closer and closer to the car's limits. Learning about and practicing "corner exit car control" will be the logical next step.

On the racetrack, you can't have been practicing the previous two subjects without using the brakes at the entry of the corners, and just because we leave the concentration on the subject for last doesn't mean that you shouldn't use the brakes. It's just that, when it comes to lowering lap time, mastering braking yields smaller lap time increments than developing the first two skills.

Still, once you've mastered the line and corner exit car control you'll find that the skill of transitioning the car from the straight to the throttle application point is the skill that separates the fastest from the fast. Braking and Entering will be our final area of concentration in dealing with the process of turning a lap in the minimum amount of time.